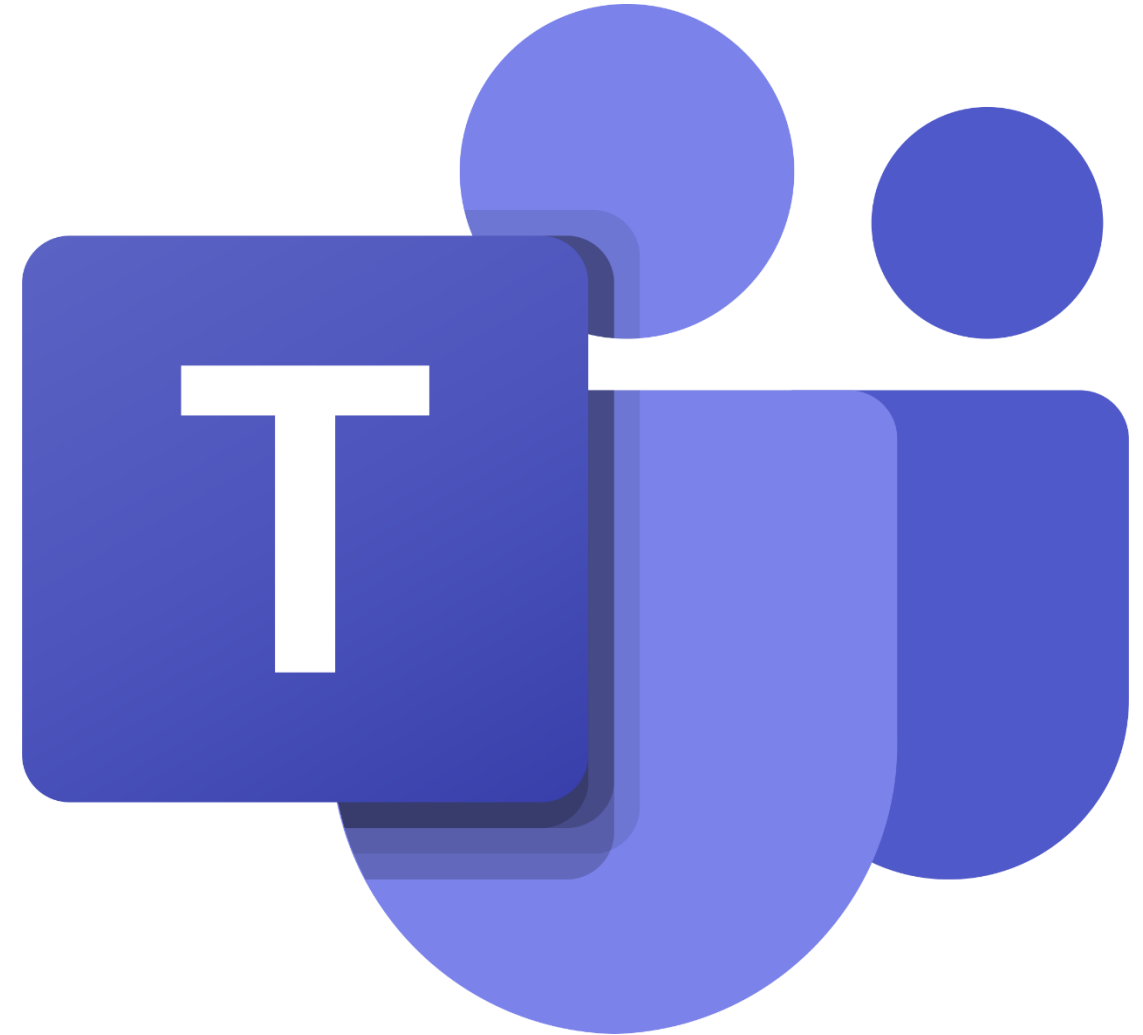


Embedded Software Engineering

Simon Künzli, Tobias Welti
HS 2025

Organizational Info: Passive Streaming on Teams

- **On request, we will use passive streaming for lecturers**
- **Link to team can be found on Moodle**
- **Code to join the ESE team on MS teams: yywj31v**
- **Register now!!**

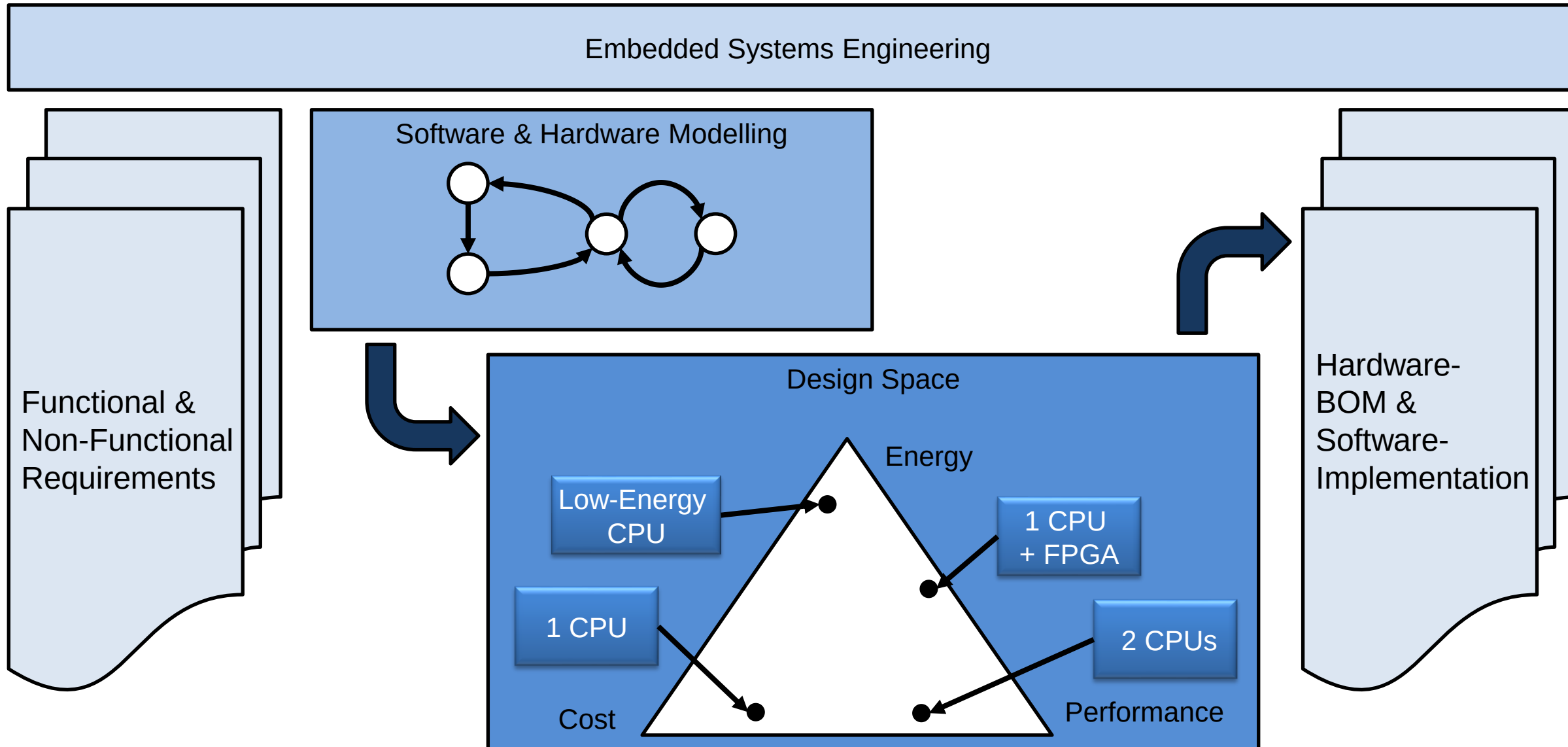


Goals

After today's lesson:

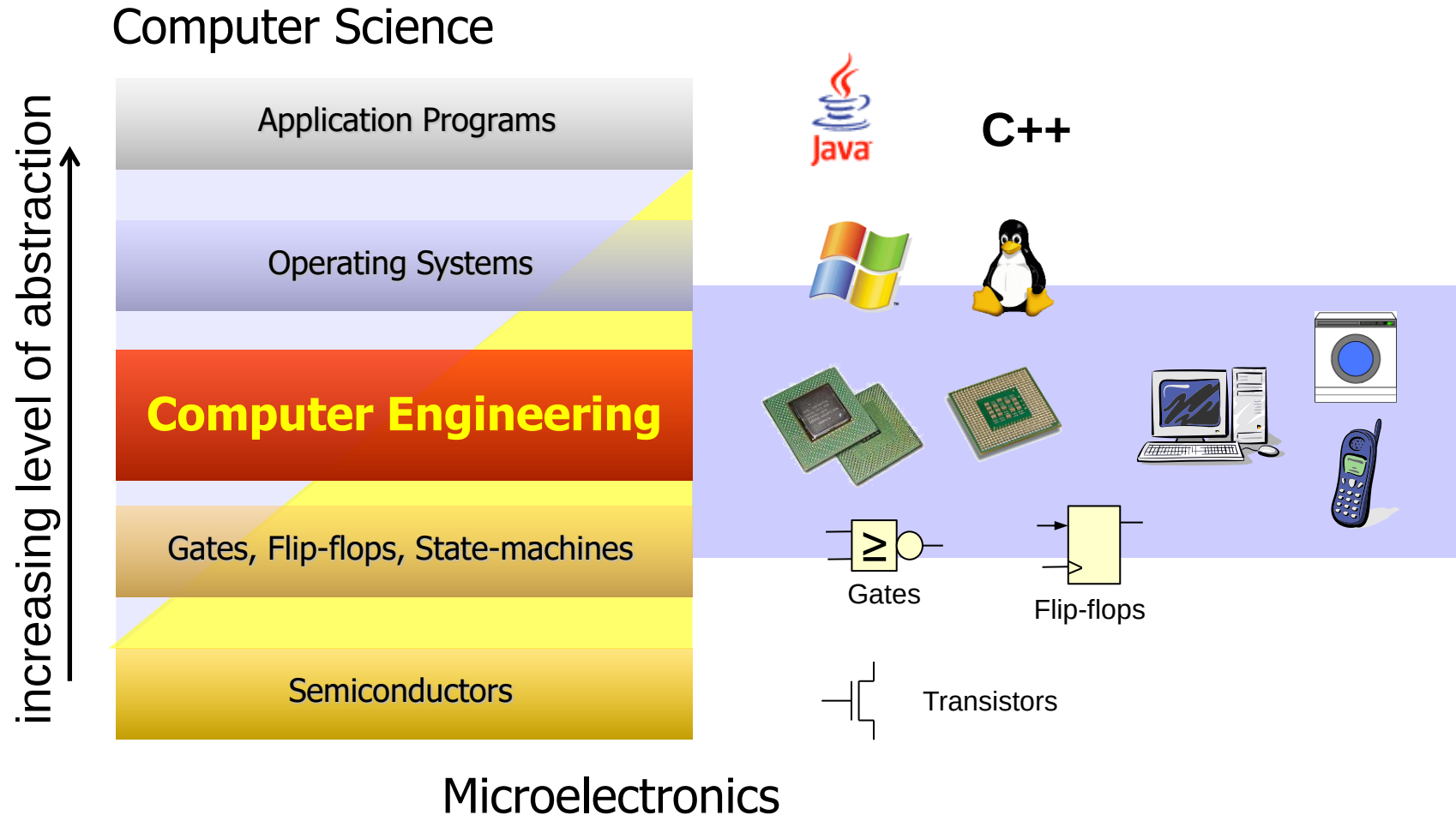
- **The students can name the main properties of an Embedded System**
- **The students know the differences between an Embedded System and a general-purpose computer**
- **The students know the main components of an Embedded System and can give examples of these components**
- **The students know what a design space is and can name several dimensions of the design space for Embedded Systems**

Advance Organizer ESE

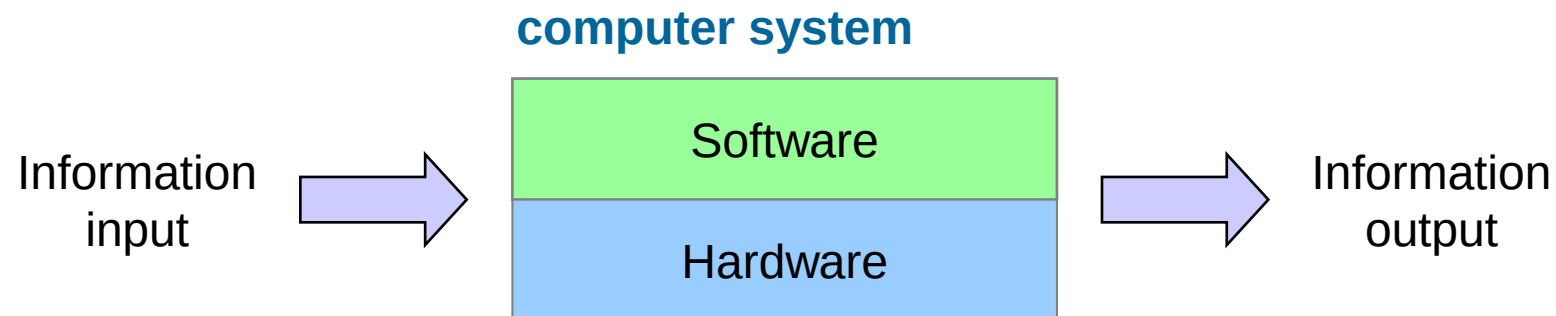


What is Computer Engineering?

Recap CT

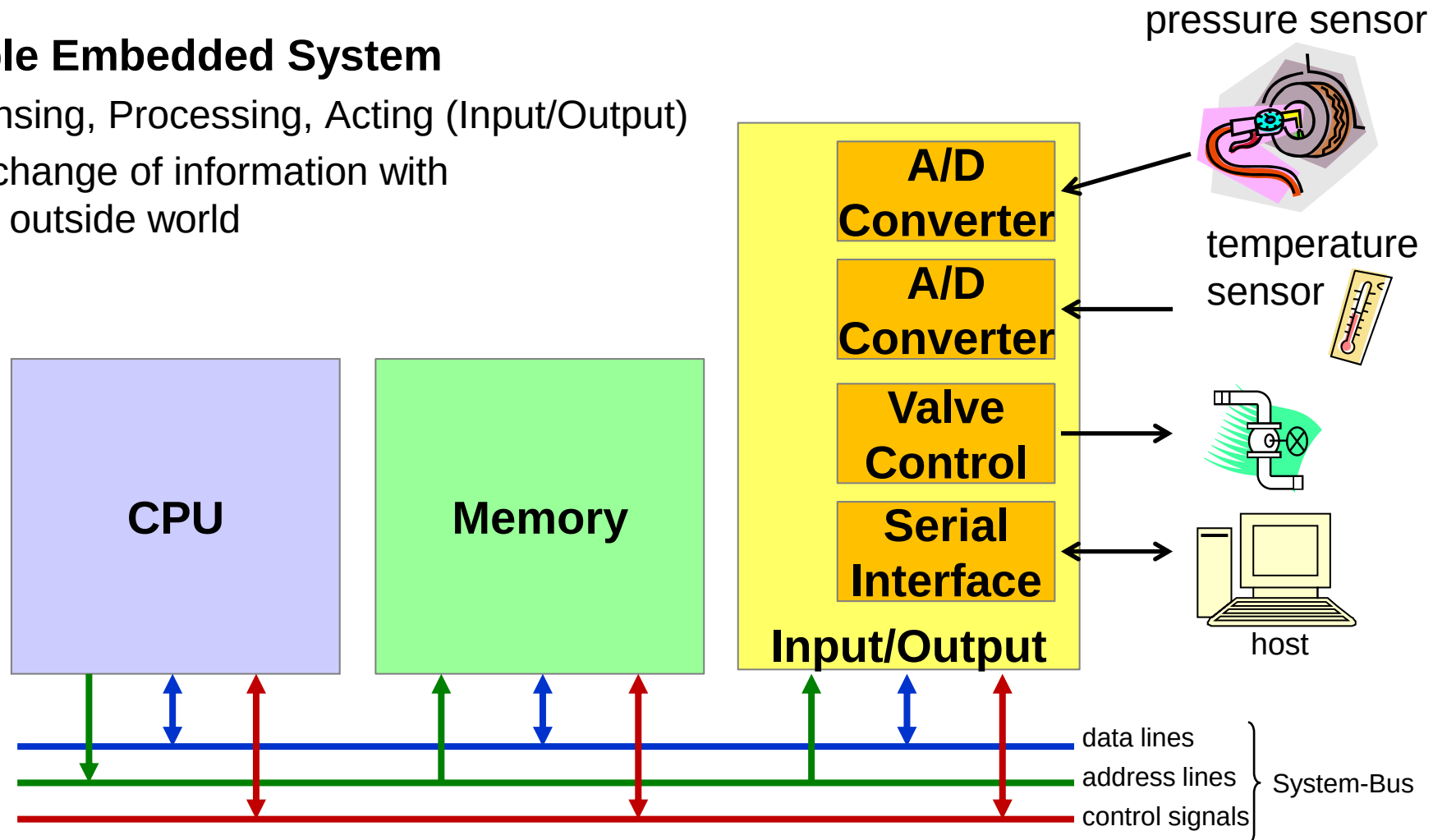


- **A computer system is a device that**
 - processes input
 - takes decisions based on the outcome
 - and outputs the processed information
- **Hardware and software work together → application**
 - often a common hardware is used for many different applications
 - application is defined by the software
 - e.g. controls for washing machines, vending machines,



■ Example Embedded System

- Sensing, Processing, Acting (Input/Output)
- Exchange of information with the outside world



Properties of Embedded Systems

■ Embedded Systems

- ES are often part of a larger system
- ES serve to control devices, facilities, processes
- ES often build wireless sensor networks (WSN)

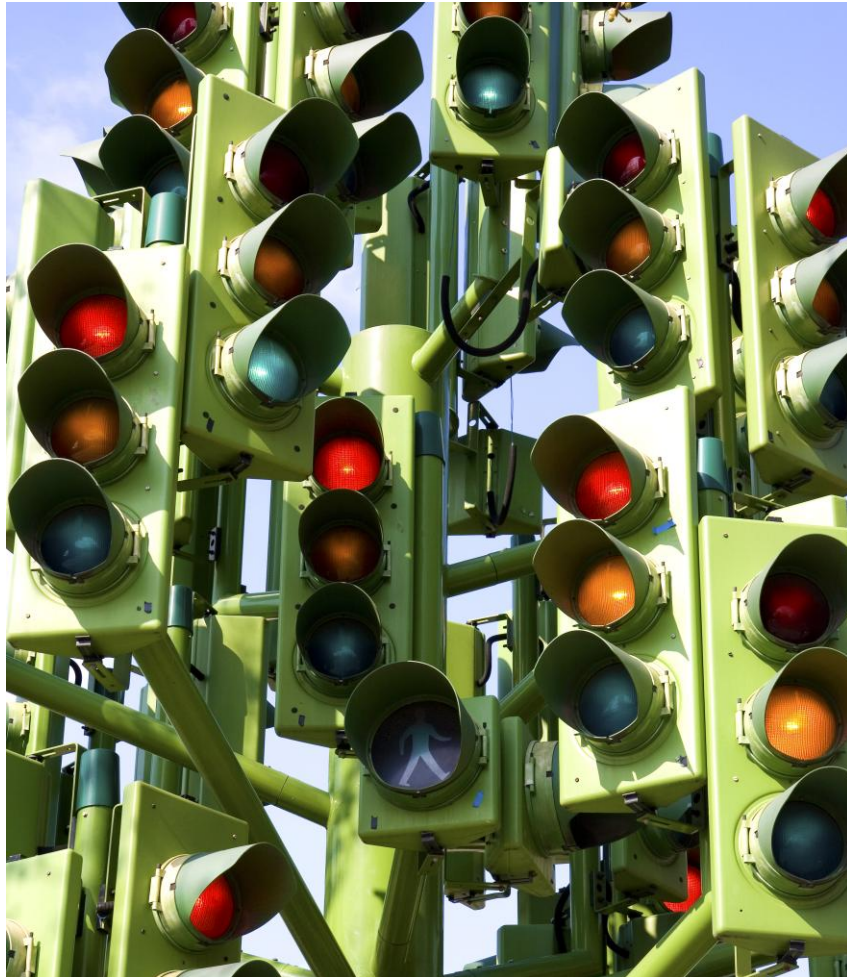
Properties of Embedded Systems (II)

- Embedded Systems must include a Processor Engine (e.g. Microcontroller)
- Embedded Systems May Come without a User Interface (or only a minor UI)
- Embedded Systems usually are designed for a purpose
- Embedded Systems often use a Real-Time Operating System
- Embedded Applications are Built-in (nothing to be added at a later stage)
- Embedded Systems are resource-limited (energy, memory, etc.)

Embedded System Projects

- **During the overall ESE module, we will work with 3 specific Embedded Systems:**
 - Traffic Light Control System
 - Washing Machine Control System
 - AES Encryption/Decryption System

Example 1: Traffic Light Control System



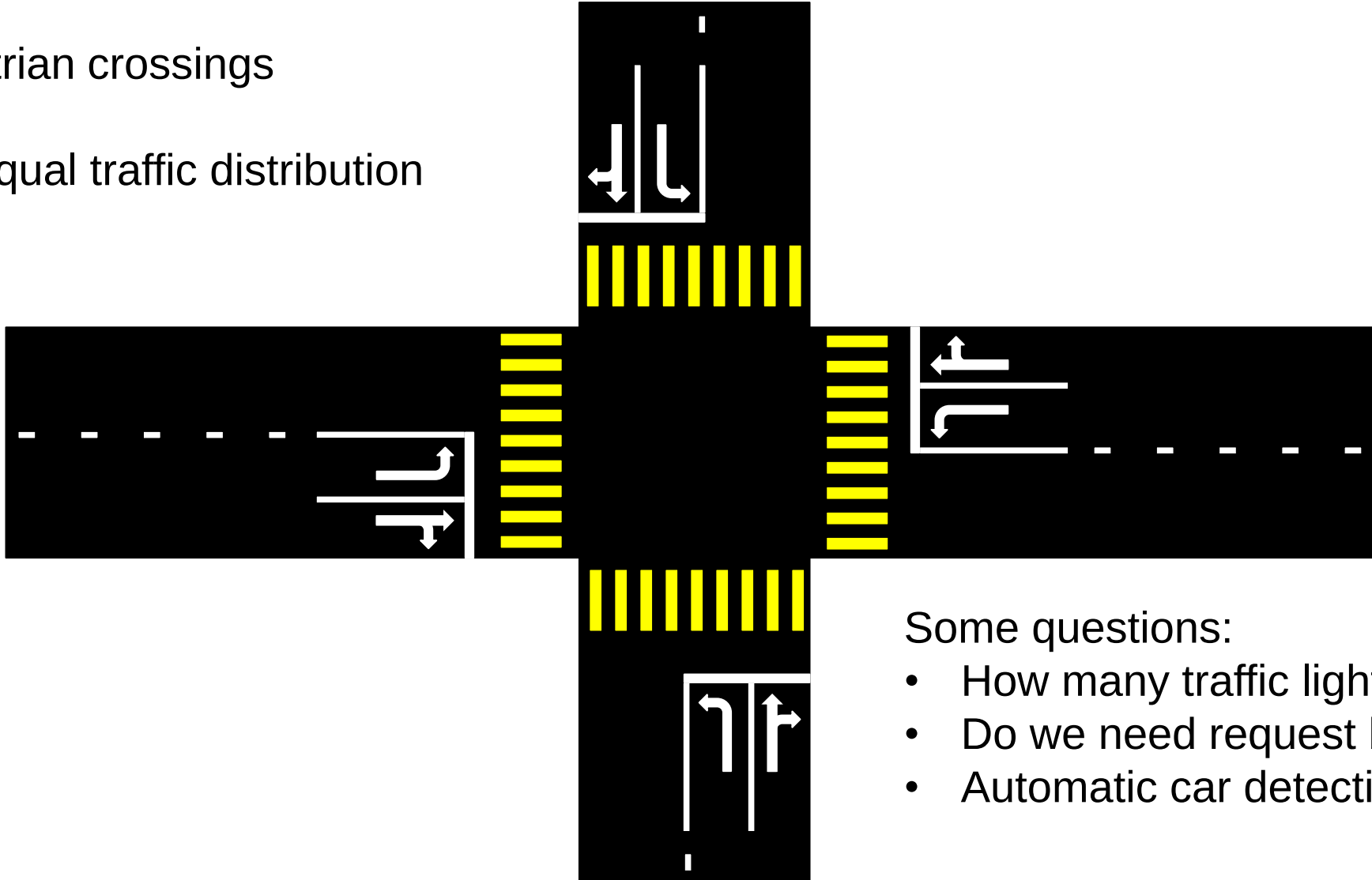
Source: Colourbox.de



Source: Colourbox.de

Example 1: Overall Situation

- Pedestrian crossings
- Cars
- (Un-)Equal traffic distribution



Some questions:

- How many traffic lights?
- Do we need request buttons?
- Automatic car detection?

Example 2: Washing Machine

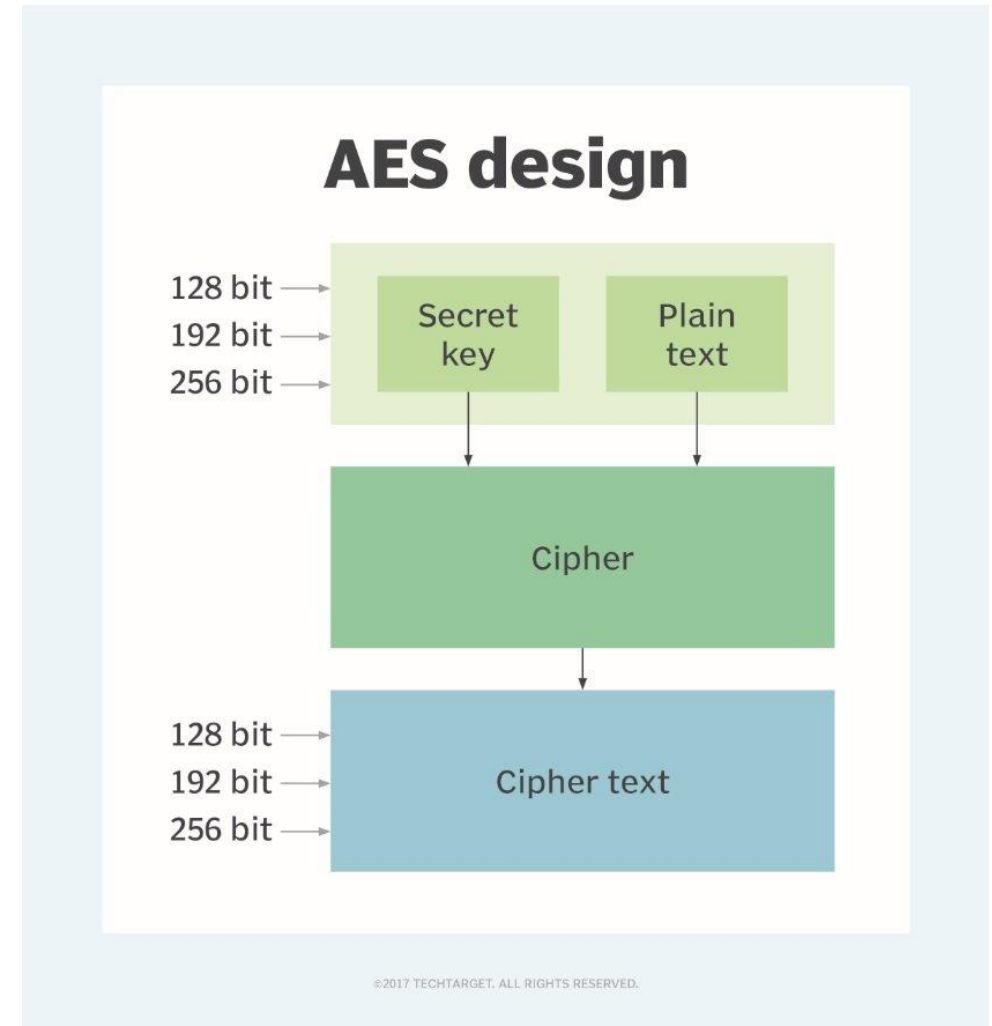
- **Control System for Washing Machine**
- **This will be the topic of the exercises U1, U2, and U3**
- **For these 3 exercises, we will work with well-known platform: the CT1 boards**
- **Check whether you have Keil uVision still installed on your computer. There are a few computers in TE 507 with uVision installed**



Source: Colourbox.de

Example 3: AES Encryption/Decryption

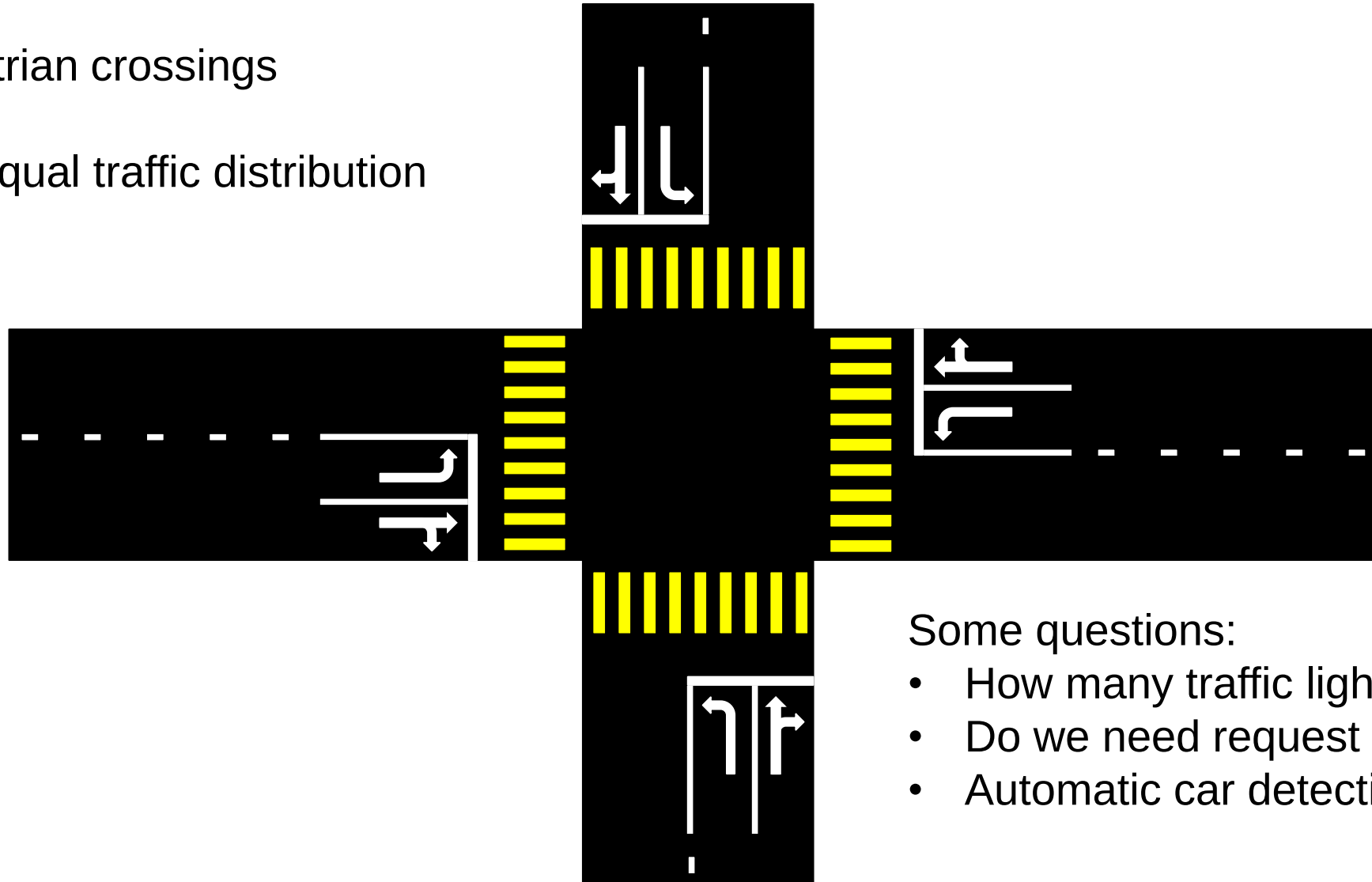
- For the practical exercises P1-P4 we will focus on AES encryption and decryption algorithms
- Implementation on Ultra-96 board, involving different types of MCUs and/or HW accelerators



Source: Techtargt.com (accessed on September 20, 2023).

Coming back to Example 1...

- Pedestrian crossings
- Cars
- (Un-)Equal traffic distribution



Some questions:

- How many traffic lights?
- Do we need request buttons?
- Automatic car detection?

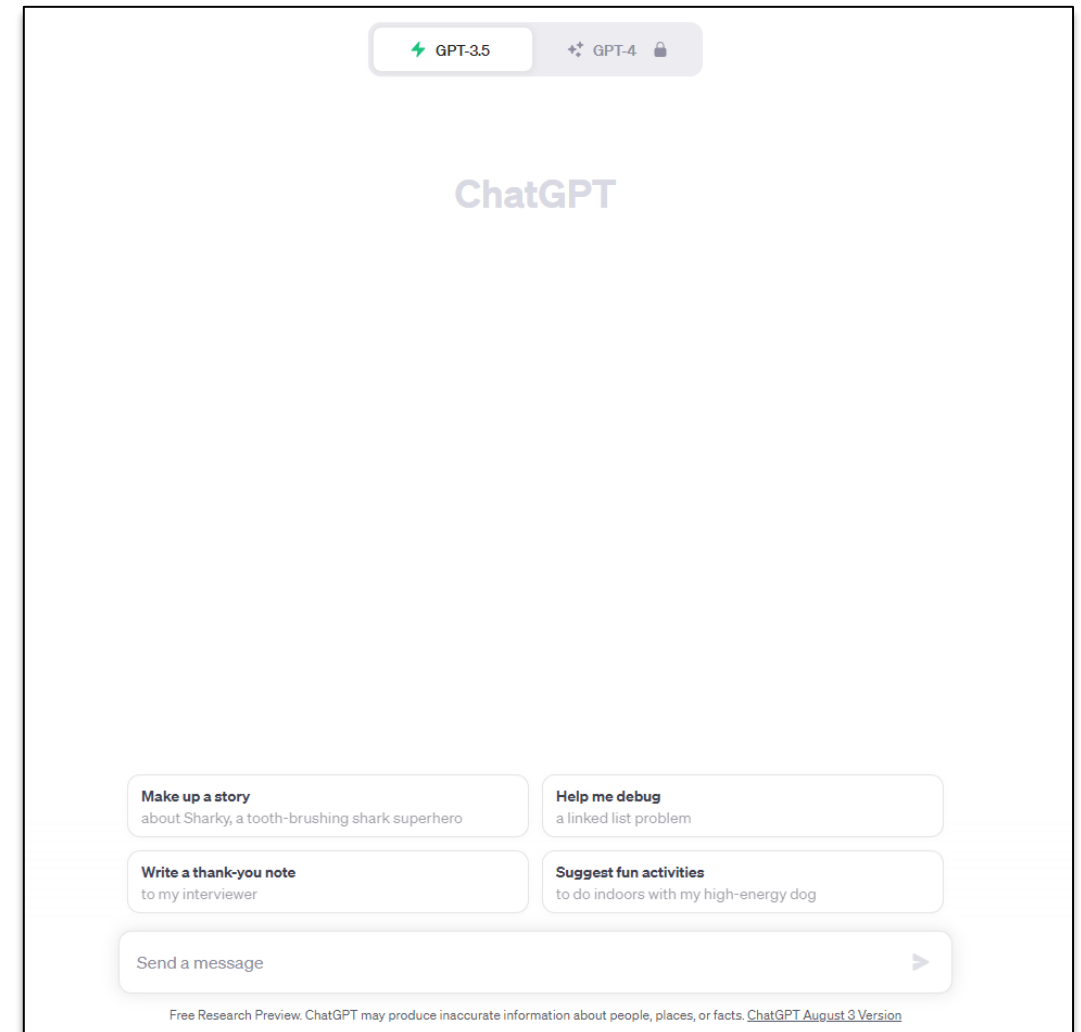
Group Discussion

- **What components do we need to build a traffic light control system?**
- **More specifically: What electronic components do we need to build the control system?**



Let's ask for a different opinion...

- **Can ChatGPT help?**
- **Revise your proposals with the additional information gained by ChatGPT**



Design Space for Embedded Systems

■ HW: Options to choose from:

- Type of MCU (e.g., ARM-M0+, ARM-R5, RISC-V, TI MSP430, etc.)
- Number of MCUs (1, 2, ...)
- Cache System (size, type, # levels)
- Hardware Accelerators (special purpose, FPGA)
- Interconnect between MCUs (bus systems, bus widths)
- Sensor Types (analog vs. digital sensor)
- ...

■ SW: We have the flexibility...

- ... of binding SW Blocks on different HW components
- ... of using operating system support
- ... of coding bare-metal applications

Eingesetzte Hardware



■ Ultra96-V2 Single Board Computer

- Wi-Fi, Bluetooth, SD-Card Slot and various I/O's
- Zynq UltraScale+ EG MPSoC
 - Quad-core Arm Cortex-A53 based Application Processing Unit (APU)
 - Dual-core Arm Cortex-R5 based Real Time Processing Unit (RPU)
 - Arm Mali-400 MP2 based Graphics Processing Unit (GPU)
 - Configuration Security Unit (CSU)
 - Programmable Logic (FPGA)
- Custom PetaLinux
 - IDE: VS Code on your laptop, cross-compilation for Ultra96
 - DHCP server for easy ssh connection over Ethernet

FPGA

XCZU3EG-1SFVC784I

Processors

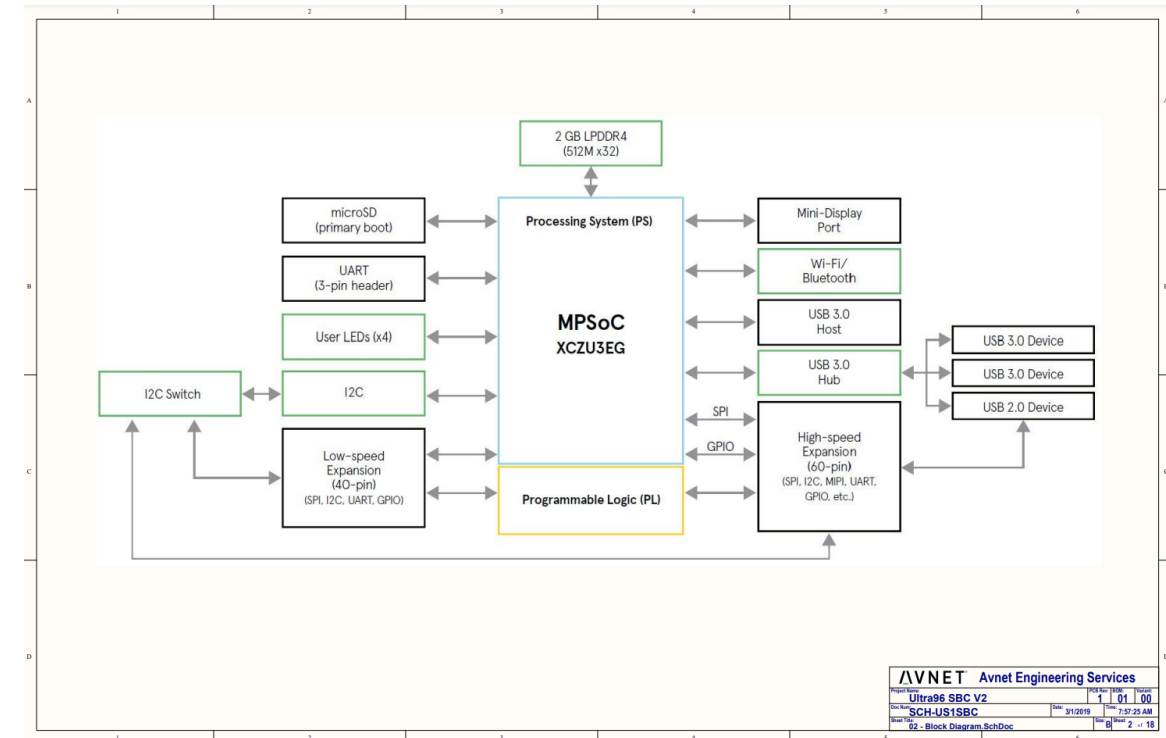
4x ARM Cortex-A53,
1.2GHz
2x ARM Cortex-R5,
500MHz

GPU

ARM Mali-400 MP2

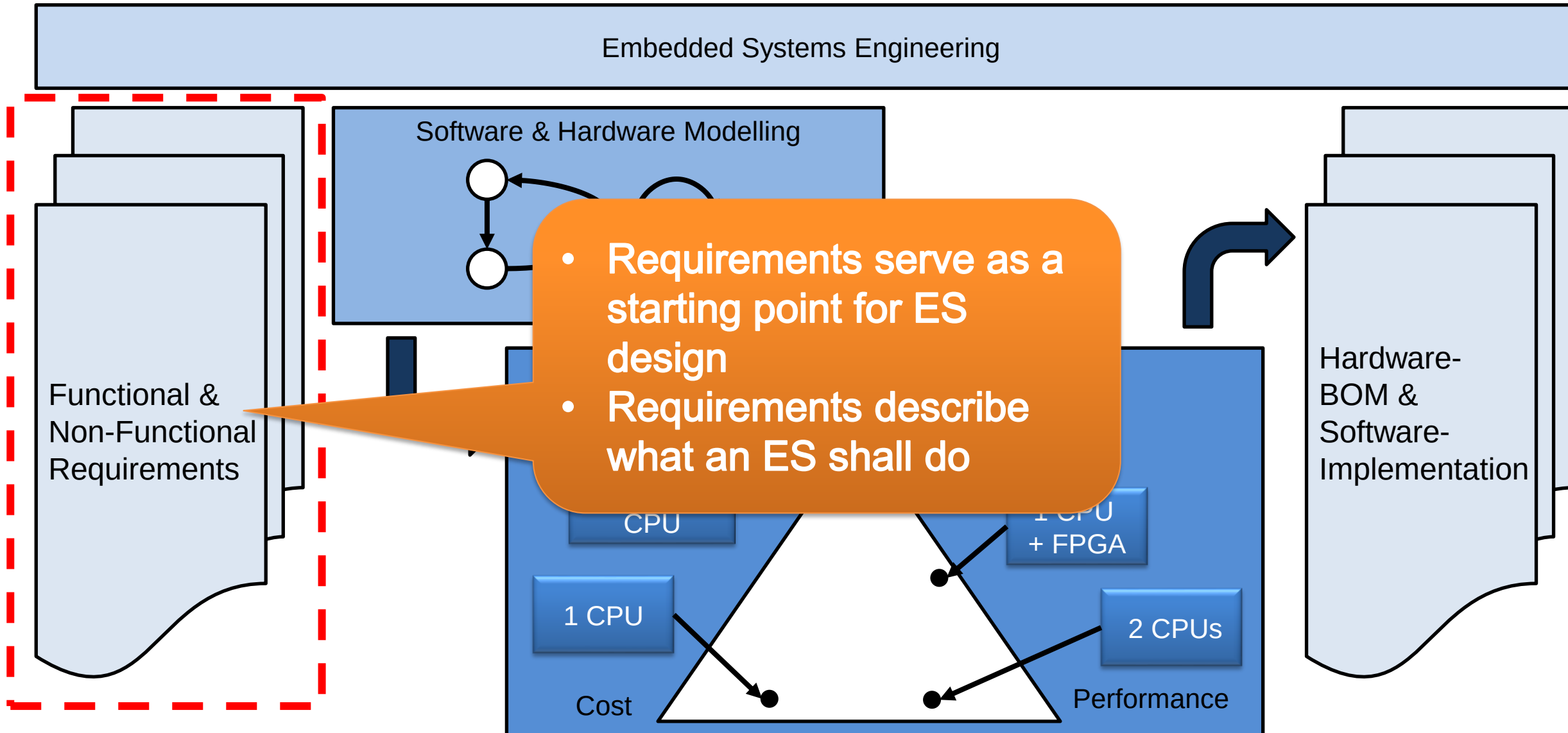
RAM

LPDDR4, 2GB, 32bit Bus,
1066 Mbps

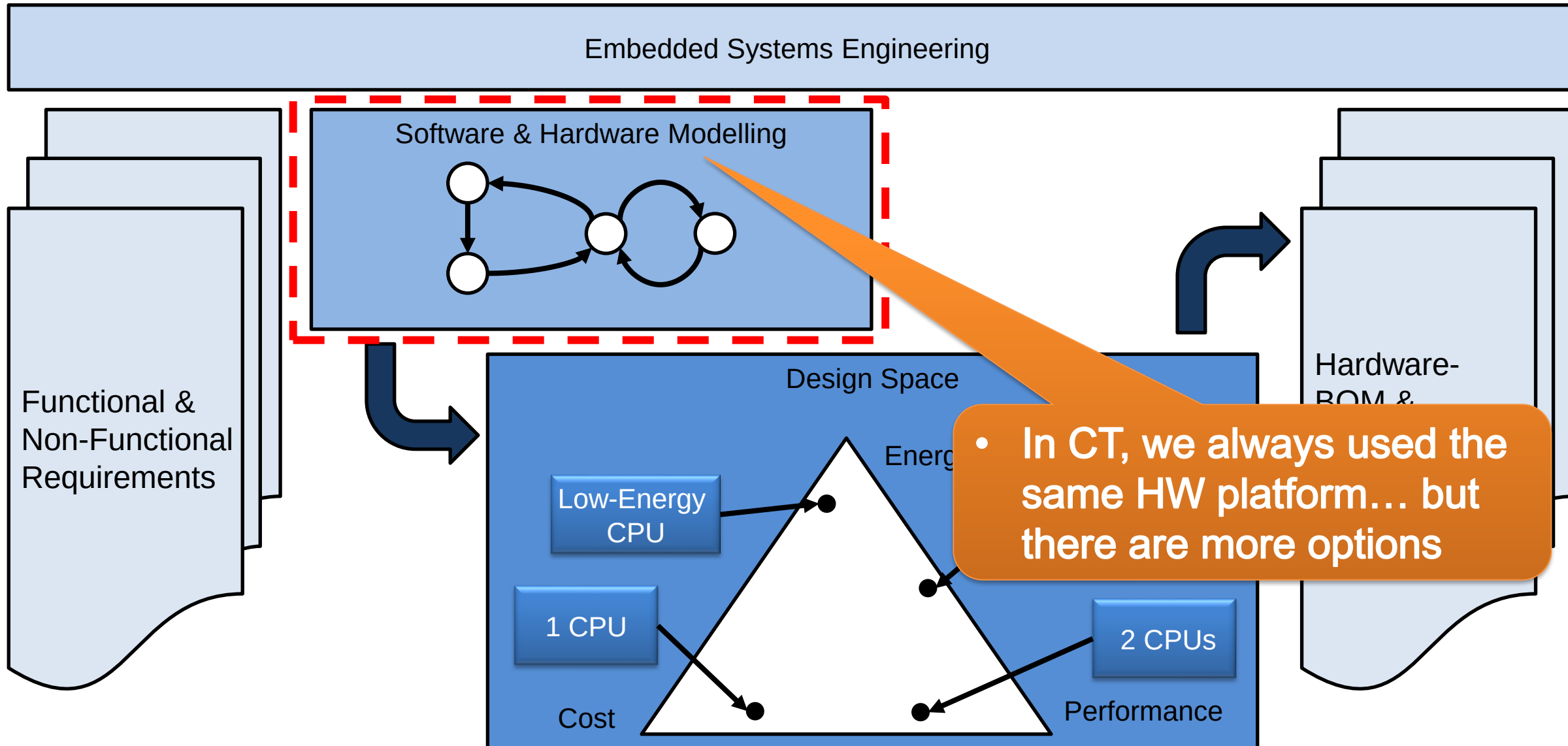


So many options...

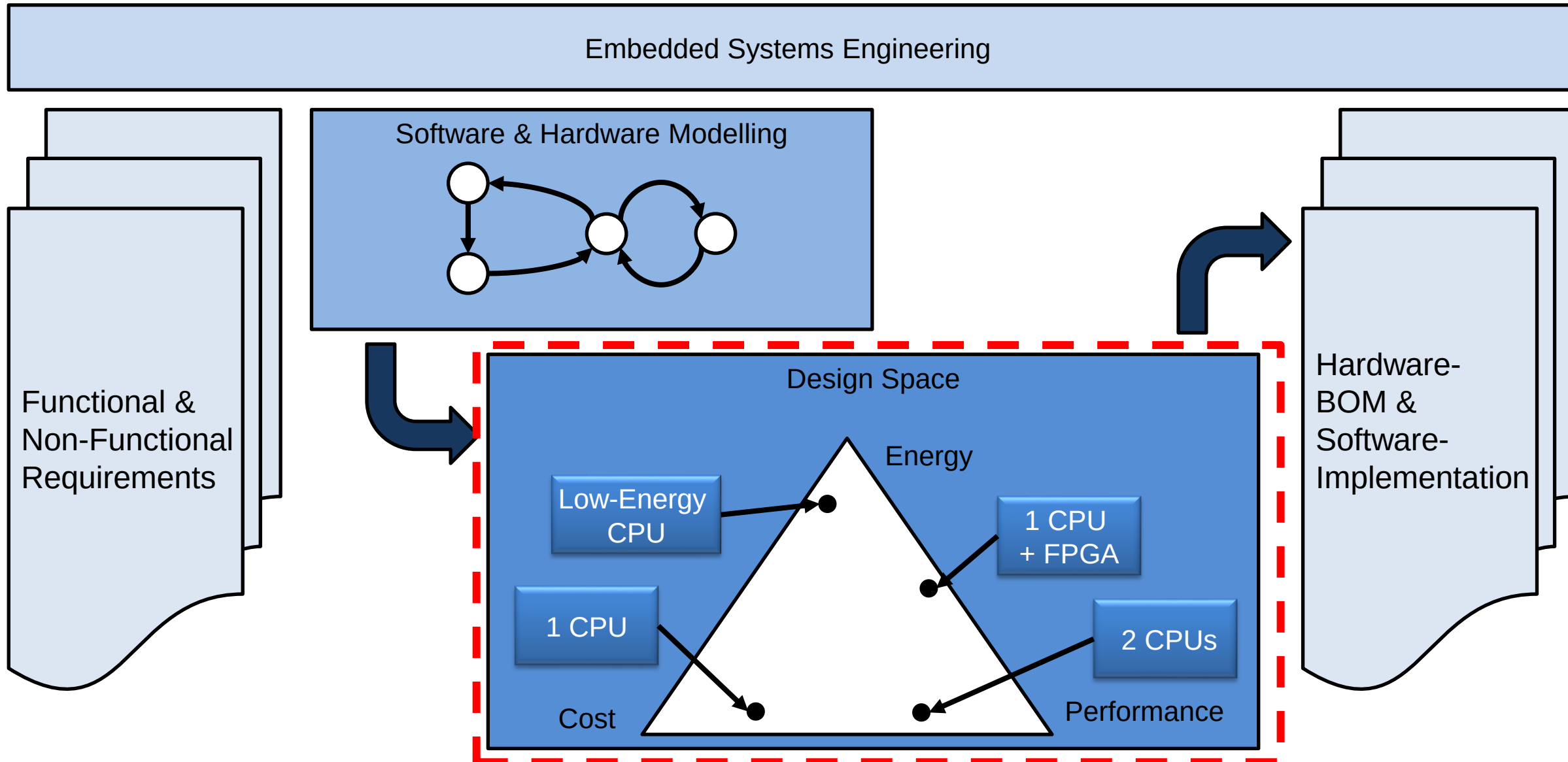
Advance Organizer ESE



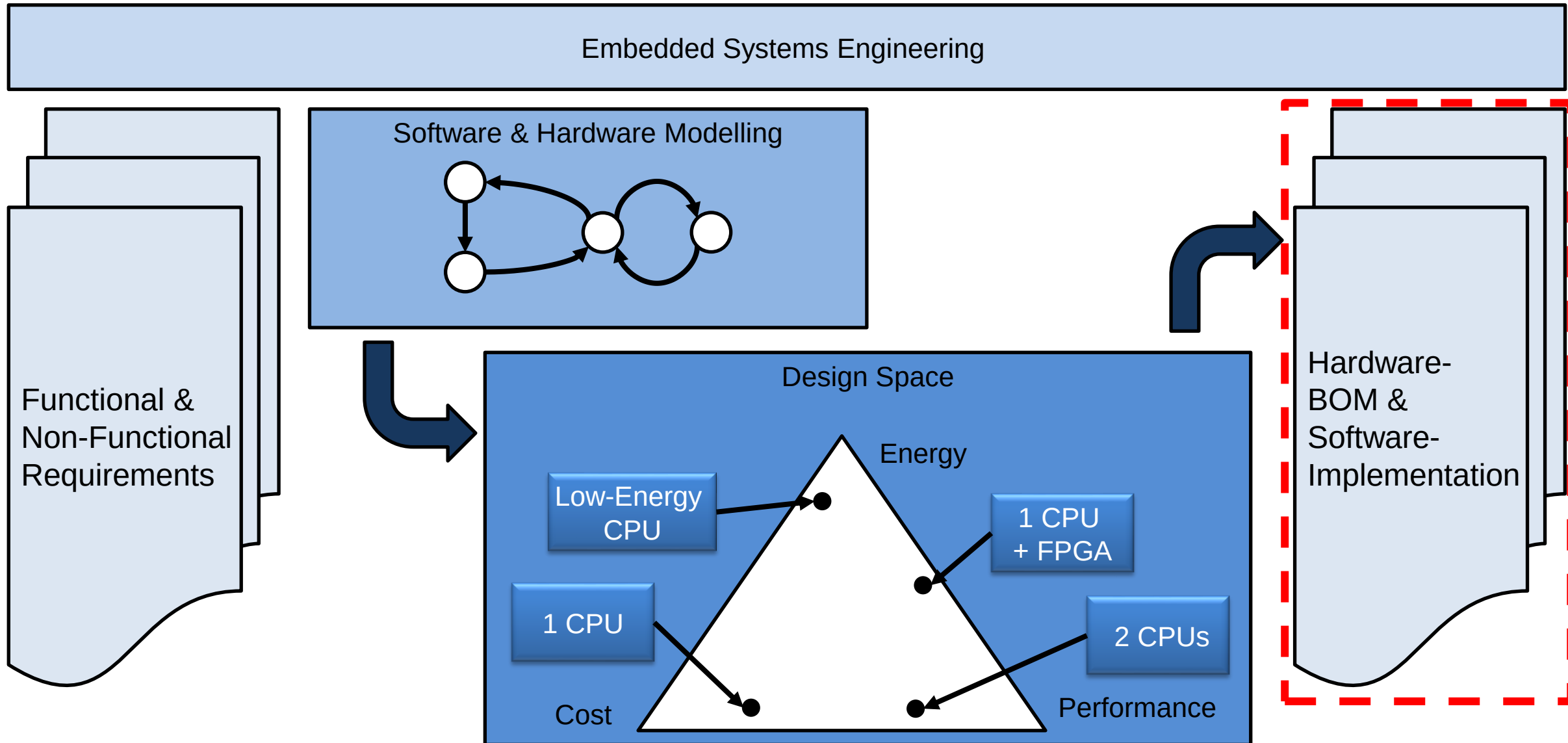
Advance Organizer ESE



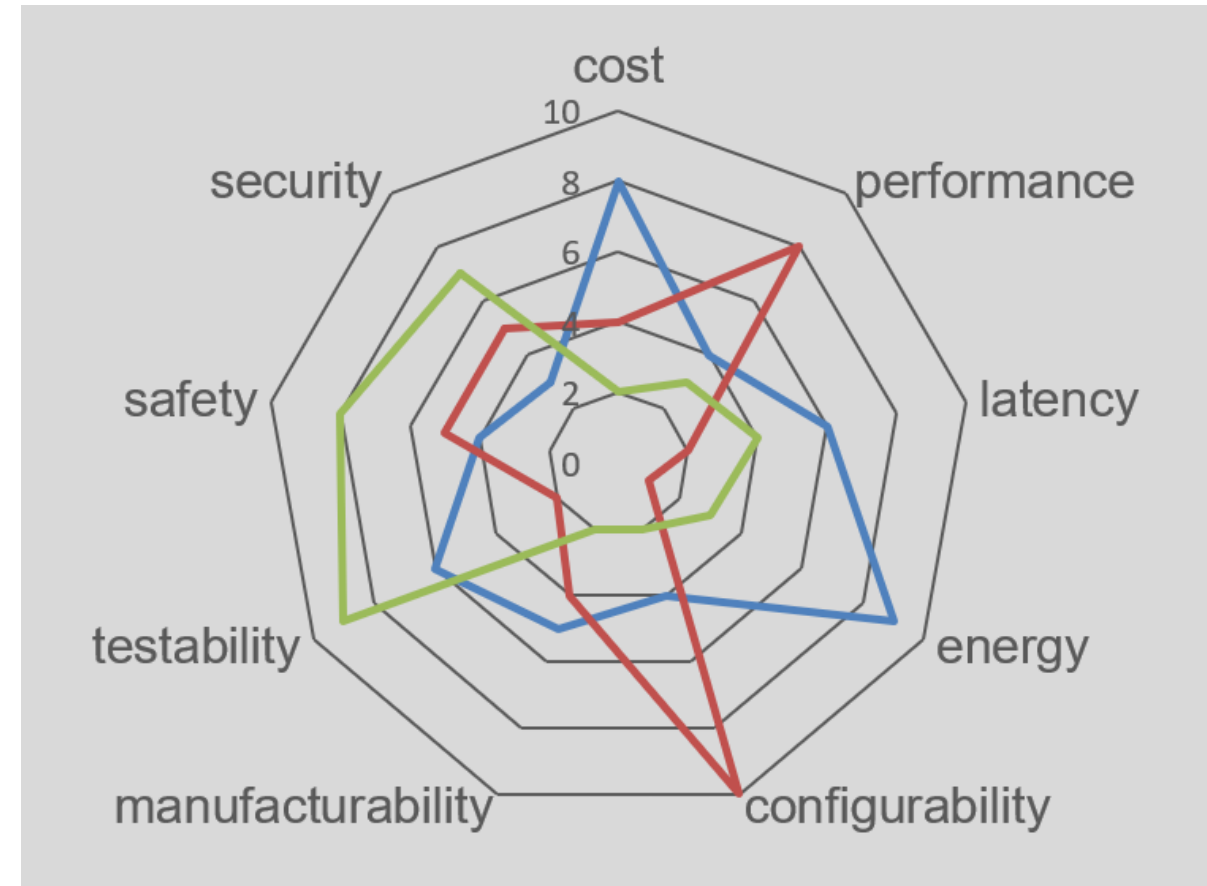
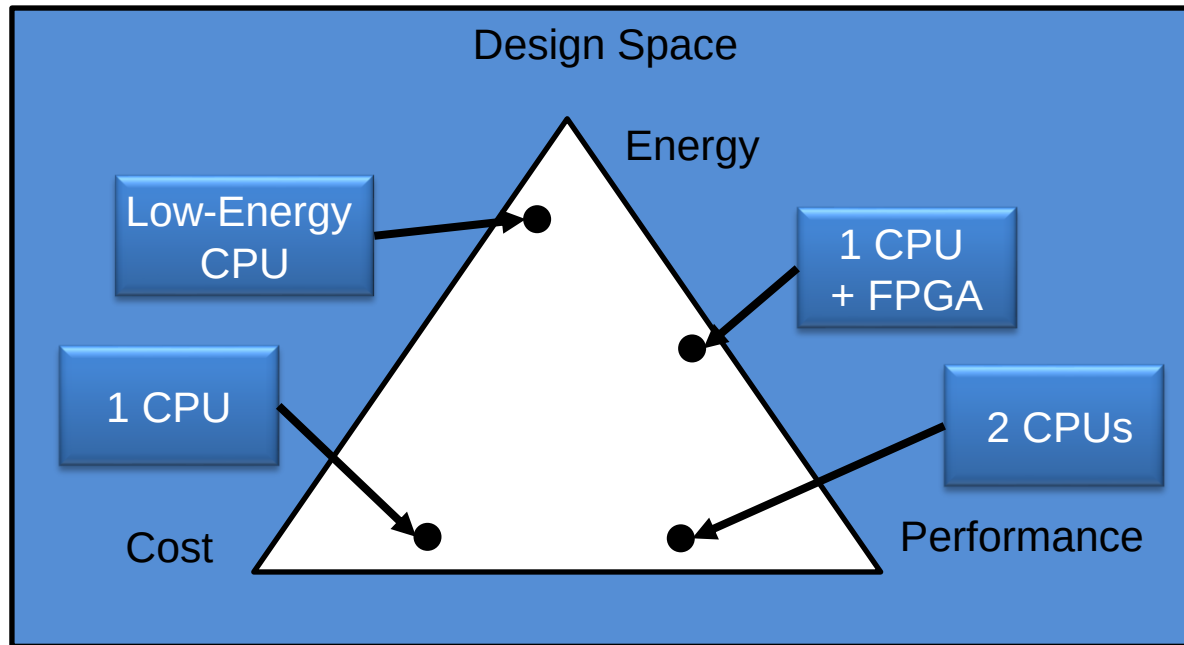
Advance Organizer ESE



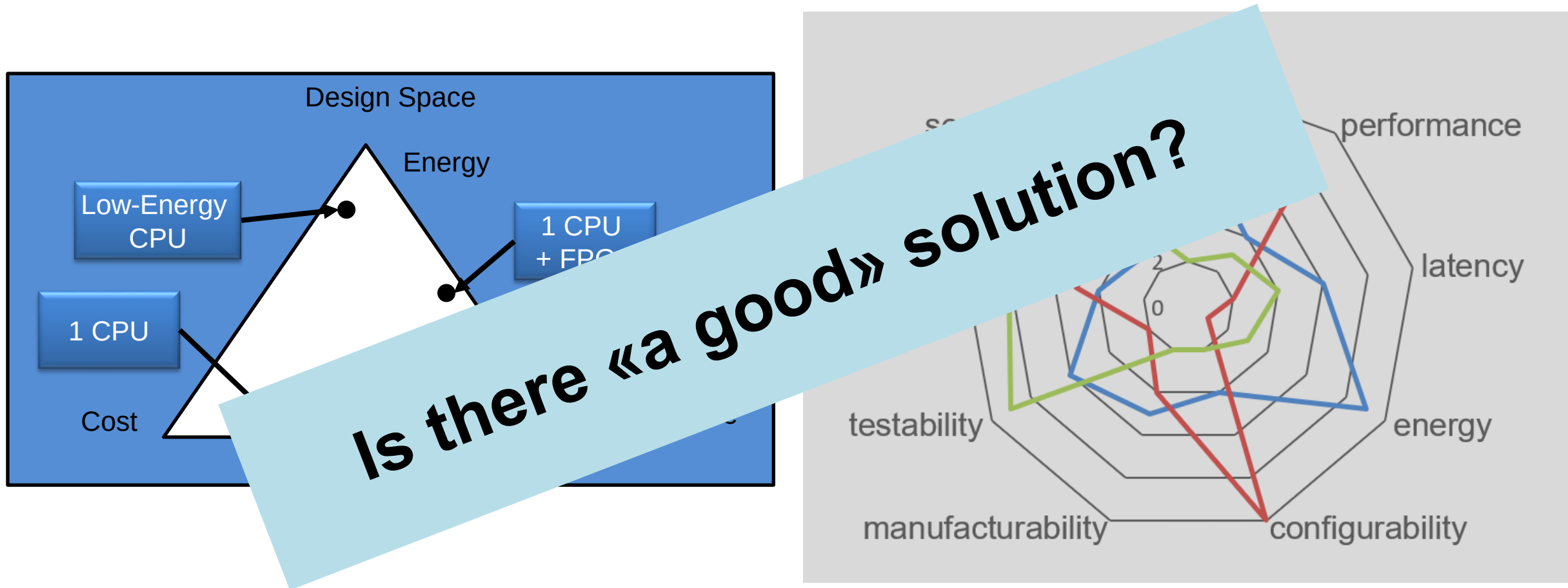
Advance Organizer ESE



Properties of Embedded Systems



Properties of Embedded Systems



Summary

- **In the Embedded Systems world, there is no single solution to a problem**
- **HW and SW architectures (may) influence each other**
- **We need a systematic approach to Embedded System Design**
- **In the following weeks, you will learn:**
 - Theory and principles of the analysis and design of Embedded Systems
 - Practical aspects of Embedded System design, such as HW selection, software design, and binding of HW and SW